

BlueKoi: Agility through unpowered turns in a fish-inspired robot

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Abstract—Bio-inspired, fish-like robots are promising platforms for collecting data within sensitive environments, due to the safer nature of oscillatory flapping propulsion. However, current fish-inspired robots often cannot match the maneuverability of their biological counterparts. This study aims to close the gap by expanding on our previous work BlueKoi, a koi- and tuna-inspired robot that captures the benefits of body flexibility with only two actuators. We introduce a rapid maneuver for angular displacements. Through experimental data in a large pool and building upon our simplified model, we demonstrate the potential for BlueKoi as a capable platform for marine operations.

Index Terms—bio-inspired, robotics, ocean vehicles

I. INTRODUCTION

Coral reefs, seagrass beds, and kelp forests are critical marine environments that demand frequent and widespread field observations, particularly data that requires close-range monitoring. Propeller-based craft have greatly expanded data collection capabilities, yet disruptive propeller wash limits how closely they can approach animals and delicate natural structures [1]. Bio-inspired, fish-like robots offer a promising alternative, as their oscillatory flapping motion has a lower risk of being destructive, posing less danger to surrounding organisms and habitats. Despite recent advancements, however, robotic fish remain considerably less maneuverable than their biological counterparts. This gap stems in part from morphological constraints: robots must house their power sources, electronics, and actuators within a rigid body, reducing the overall flexibility that is central to fish-like agility [2].

To bridge this, our previous work introduced BlueKoi [3] (Fig 1C), a koi- and tuna-inspired robot designed for real-world field environments. Despite containing only two actuators, BlueKoi executes maneuvers that exploit full-body flexibility while remaining mechanically simple. By contrast, other fish-inspired robots that rely on body bending for maneuvering require multiple actuators linked in series, an arrangement that may limit maximum operating frequency, and thus speed [4].

This study expands upon our previous work on BlueKoi by presenting a novel 2D maneuver, inspired by fish “c-start” [5], that fully utilizes BlueKoi’s unique morphology to execute highly tunable turning. We experimentally quantify the turning

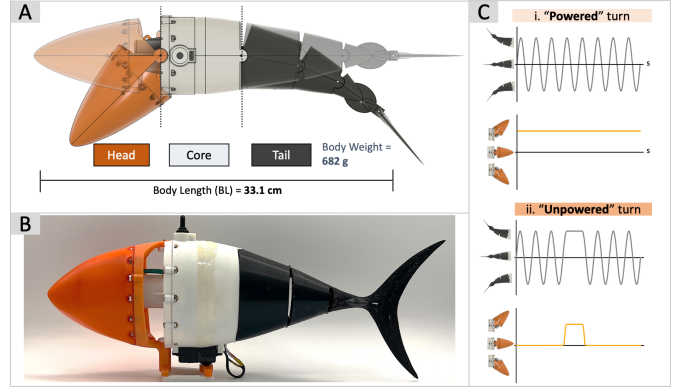


Fig. 1. (A) The design of the robot, demonstrating maximum positions for head and tail. (B) The flat plate model of the robot, as described in [3] (C) Photo of robot. (D) Head and tail positions for a **powered** turn (i) and **unpowered** turn (ii)

behavior and the driving variables, and we show that these results can be reproduced using a modified version of our previous data-driven model, demonstrating BlueKoi’s potential as a low-cost and scalable platform for ocean exploration.

II. BLUEKOI DESIGN

Fish-inspired robots that rely solely on the caudal fin for maneuvering are non-holonomic in their motion. As a result, turning in 2D can be non-trivial. One approach is to employ an asymmetric tail beat [2]; however, this can be challenging to implement mechanically and may limit the total generated thrust from the tail. Instead, BlueKoi performs turns by leveraging its unique ability to rotate both its head and tail to break the body’s spatial symmetry. One maneuver the BlueKoi is capable of is a “**powered**” turn, wherein the head is actuated to one side to steer the robot while the tail continues to flap symmetrically (Fig. 1D(i)). This creates unequal drag across the left and right sides of the robot, resulting in a wide circular turn [3]. However, the powered turn results in a large minimum turn radius and is unsuitable for rapid maneuvers such as last-second collision avoidance.

A new and more agile maneuver we demonstrate in this study is the “**unpowered**” turn, during which the head and tail both are rapidly actuated to one side and held in place

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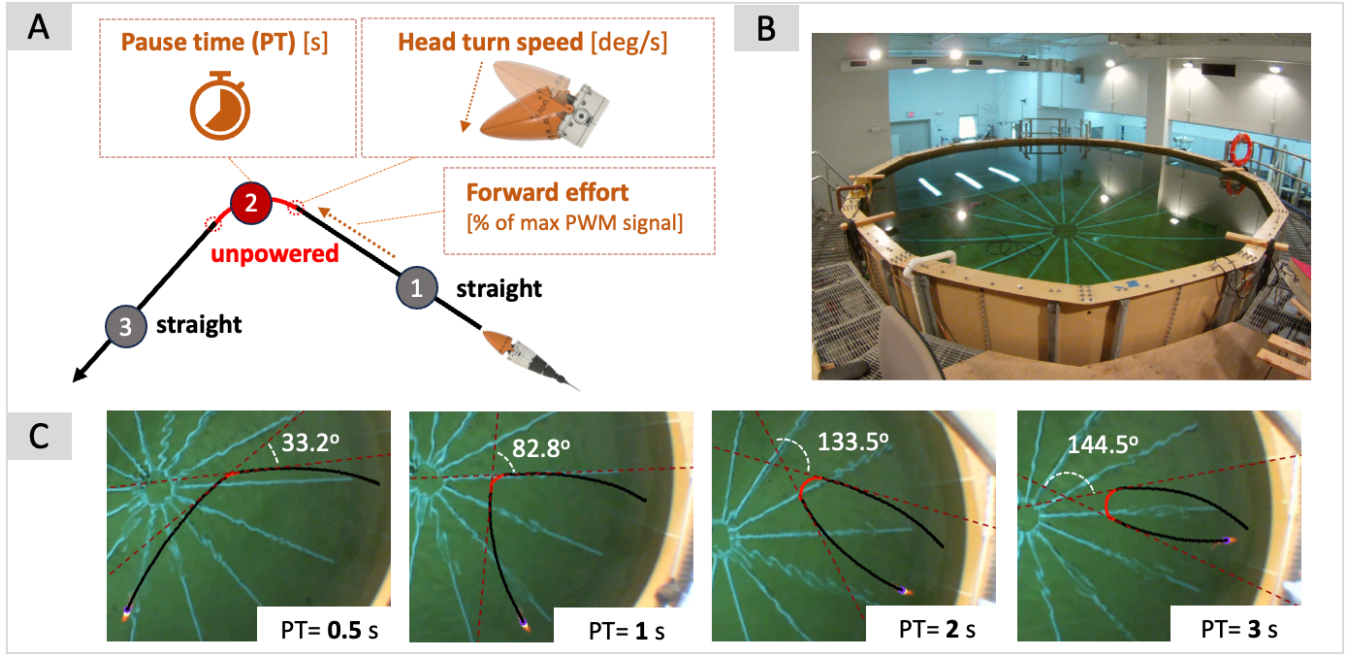


Fig. 2. (A) Example trajectories of the unpowered turning maneuver with varying pause times. To reduce angle displacement measurement error due to a curved straight line path, we take only the 0.5 seconds before and after the turn to approximate the headings. (B) The angle difference determined by the approximate changes in heading. (C) A powered turn maneuver after an initial straight motion path. (D) A trajectory containing straight forward motion, an unpowered turn, and a powered turn. (E) The unpowered turn, with a pause time of 2 seconds, on a shallow beach in New Jersey

Fig. 3. (A) Example trajectories of the unpowered turning maneuver with varying pause times. To reduce angle displacement measurement error due to a curved straight line path, we take only the 0.5 seconds before and after the turn to approximate the headings. (B) The angle difference determined by the approximate changes in heading. (C) A powered turn maneuver after an initial straight motion path. (D) A trajectory containing straight forward motion, an unpowered turn, and a powered turn. (E) The unpowered turn, with a pause time of 2 seconds, on a shallow beach in New Jersey

Fig. 4. (A) Example trajectories of the unpowered turning maneuver with varying pause times. To reduce angle displacement measurement error due to a curved straight line path, we take only the 0.5 seconds before and after the turn to approximate the headings. (B) The angle difference determined by the approximate changes in heading. (C) A powered turn maneuver after an initial straight motion path. (D) A trajectory containing straight forward motion, an unpowered turn, and a powered turn. (E) The unpowered turn, with a pause time of 2 seconds, on a shallow beach in New Jersey

(Fig. 1A, D(ii)). The sudden, rapid change to an asymmetric “c” shape coupled with an abrupt and loss of thrust results in a momentum-driven tight angular displacement, or partial turn.

III. EXPERIMENTS

Methodology. In order to better understand the characteristics and limits of the unpowered turn, we quantify the maneuver through trajectory tracking experiments in a large pool. The trajectories are executed in three phases: 1) straight forward swimming, 2) unpowered turning, 3) straight forward swimming in a new direction (Fig. 4A). The initial forward swimming generates momentum for the unpowered turn. During forward motion, the head is held centered while the tail provides thrust. During the turn phase, both the head and tail are held at their furthest possible angle from center. The time spent in the turn phase, known as the **pause time** (PT), is varied across 0.5, 1, 2, and 3 seconds (Fig. 4A), after which the robot resumes forward swimming.

Preliminary Results. From these experiments, we demonstrate a relationship between the PT and the total angular displacement achieved during the turn (Fig. 4B). We can quantify the turning by determining an approximate change in

heading between the two straight motion paths. The straight line paths appear slightly curved due to a combination of camera distortion and small bias in that particular robot’s steering. Generally, the longer the pause is, the more angular displacement the robot achieves (Fig. 4B). However, past a certain point, increasing the pause time does not increase the overall angular displacement due to the body’s deceleration from moving through a fluid. Nevertheless, when compared with a powered turn (Fig. 4C), it is apparent that the unpowered turn is capable of a much more immediate change in heading.

IV. ONGOING WORK

Our ongoing work builds upon the preliminary results by performing a parametric analysis investigating the impact of other control variables on the turning performance, including the speed during the initial forward motion phase and the angle of the head during the turning phase. Furthermore, we explore how combinations of maneuvering strategies can achieve more complex trajectories (Fig. 4D) that would be relevant for obstacle avoidance within a prescribed larger area in a real-world environment (Fig. 4E).

A hydrodynamic model allows us to predict the trajectory more accurately and to build a foundation for controls. The model described in the previous paper [3] is a simplified, flat-plate representation of the robot (Fig. 1B), and utilizes the forces acting on the bodies to describe a trajectory. One key development in this study will be to modify the simulation to represent the unpowered turn and then to compare these results to the simulated powered turns.

Overall, we demonstrate that BlueKoi's distinctive body-bending configuration improves maneuverability and gives rise to a novel rapid unpowered turns, expanding safe deployment capability in unstructured and complex environments.

REFERENCES

- [1] L. Cai, Y. Jézéquel, T. A. Mooney, and Y. Girdhar, "Measuring and minimizing disturbance of marine animals to underwater vehicles," *arXiv preprint arXiv:2506.11335*, 2025.
- [2] Q. Zhong and D. B. Quinn, "Streamwise and lateral maneuvers of a fish-inspired hydrofoil," *Bioinspiration & Biomimetics*, vol. 16, no. 5, p. 056015, aug 2021.
- [3] I. Sha, D. B. Quinn, and R. Nagpal, "Bluekoi: Combining a tuna-inspired tail and koi-inspired body bending for maneuverability," in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 12 195–12 202.
- [4] S. Ma, Q. Zhao, M. Ding, M. Zhang, L. Zhao, C. Huang, J. Zhang, X. Liang, J. Yuan, X. Wang *et al.*, "A review of robotic fish based on smart materials," *Biomimetics*, vol. 8, no. 2, p. 227, 2023.
- [5] G. Li, U. K. Müller, J. L. van Leeuwen, and H. Liu, "Escape trajectories are deflected when fish larvae intercept their own c-start wake," *Journal of The Royal Society Interface*, vol. 11, no. 101, 2014.